

Lab - 2 : Compartment Modelling of Linear Systems

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In this lab, we numerically and analytically analyze the lake pollution model, The dynamics of the medicine in blood and in GI track with single dose of a medicine and by course of medicine with using mathematical tools and simulation software.

I. INTRODUCTION

This lab is aimed to predict different compartment modellings of Linear systems. Here, we predict how the concentration of pollution in lake will be saturated with respect to time. Also we will test the same for different situation with respect to time or initial concentration. In single dose of medicine and in course of medicine, by the dynamics of the medicine in the GI tract and in the blood stream, we will plot the saturation of medicine dose with respect to time for both the GI track and for the blood track.

II. MODEL

The concentration $C(t)$ of pollutants in a lake follows the equation

$$\dot{C} = a - bC \quad (1)$$

where $a = FC_{in}/V$ and $b = F/V$. Here C_{in} in the constant pollutant concentration of inflow into the lake, F is the fixed volumetric flow rate and V is the fixed volume of the lake (as the lake also drains out).

The concentration $C(t)$ of pollutant is given by

$$C = C_{in} + (C_0 - C_{in})e^{-Ft/V} \quad (2)$$

A single dose of a medicine is administered to a patient. The dynamics of the medicine follows the equation $\dot{x} = -k_1x$, $x(0) = x_0$ in the GI tract, and $\dot{y} = k_1x - k_2y$, $y(0) = 0$ in the blood.

The amount of drug in GI tract is given by

$$x = x_0e^{-k_1t} \quad (3)$$

and the amount of drug in blood is given by

$$y = \frac{k_1x_0}{k_2 - k_1}(e^{-k_1t} - e^{-k_2t}) \quad (4)$$

However for the same values of rate constants k_1 and k_2 the above equations differ as

$$x = x_0e^{-kt} \quad (5)$$

$$y = kx_0te^{-kt} \quad (6)$$

A course of medicine is administered to a patient. The dynamics of the medicine follows the equation $\dot{x} = I - k_1x$, $x(0) = 0$ in the GI tract, and $\dot{y} = k_1x - k_2y$, $y(0) = 0$ in the blood stream.

The amount of drug in GI tract is given by

$$x = \frac{I}{k_1}(1 - e^{-k_1t}) \quad (7)$$

The amount of drug in blood is given by

$$y = \frac{I}{k_2}(1 - e^{-k_2t}) - \frac{I}{k_2 - k_1}(e^{-k_1t} - e^{-k_2t}) \quad (8)$$

However for the same values of rate constants k_1 and k_2 the above equations differ as

$$x = \frac{I}{k}(1 - e^{-kt}) \quad (9)$$

$$y = \frac{I}{k}[1 - (kt + 1)e^{-kt}] \quad (10)$$

III. RESULTS

A. Plot the concentration of pollutants in the lake

Fig. 1 shows the concentration of pollutants in the lake over time.

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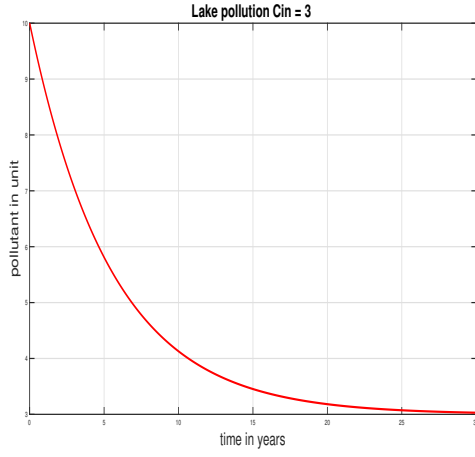


FIG. 1: Here $F = 5 \times 10^8 m^3/day$, $V = 10^{12} m^3$, $C_{in} = 3$ unit and $C(0) = C_0 = 10$ unit.

The time taken for $C = 0.5C_0$ is **2505.5 days** when C_{in} is not zero. The time taken for $C = 0.5C_0$ is **1386.3 days** when C_{in} is zero.

Fig. 2 shows the concentration of pollutants in the lake over time when C_{in} is zero.

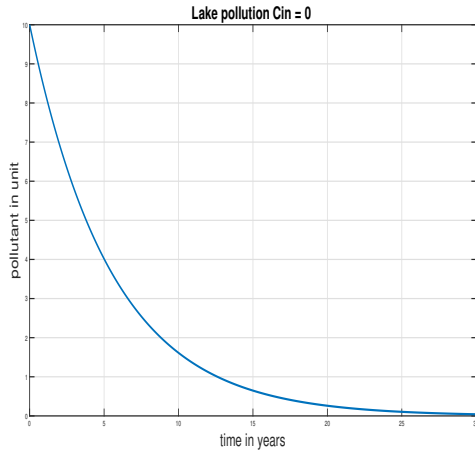


FIG. 2: Here $F = 5 \times 10^8 m^3/day$, $V = 10^{12} m^3$, $C_{in} = 0$ unit and $C(0) = C_0 = 10$ unit.

B. Plot the amount of drug in a single dose of medicine

Fig. 3 shows the amount of drug in GI tract and blood stream with respect to time.

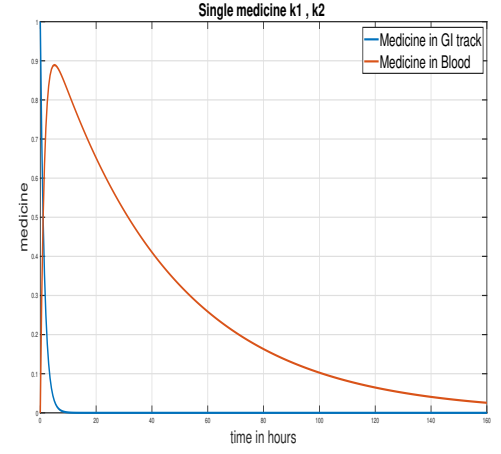


FIG. 3: $k_1 = 0.6931 hr^{-1}$, $k_2 = 0.0231 hr^{-1}$, $x_0 = 1 unit$, $y_0 = 0 unit$

The peak value of $y(t)$ is **0.889** units at $t = 5.1$ hours.

Fig. 4 shows the amount of drug in GI tract and blood stream with respect to time when $k_1 = k_2$.

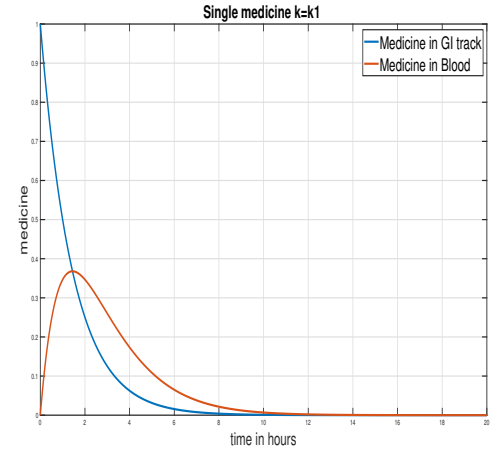


FIG. 4: $k_1 = k_2 = 0.6931 hr^{-1}$, $x_0 = 1 unit$, $y_0 = 0 unit$

Fig. 5 shows the amount of drug in GI tract and blood stream with respect to time when $k_1=k_2$.

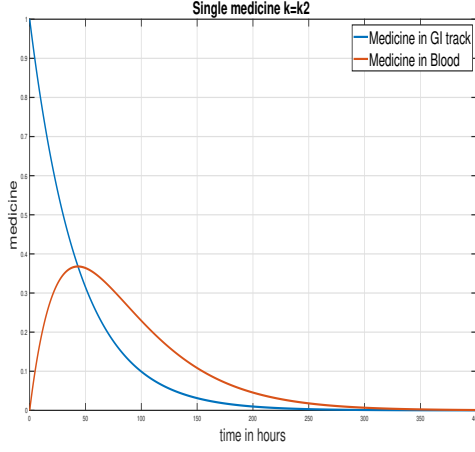


FIG. 5: $k_1 = k_2 = 0.0231hr^{-1}$, $x_0 = 1 unit$, $y_0 = 0 unit$

Fig. 7 shows the amount of medicine in blood with respect to time.

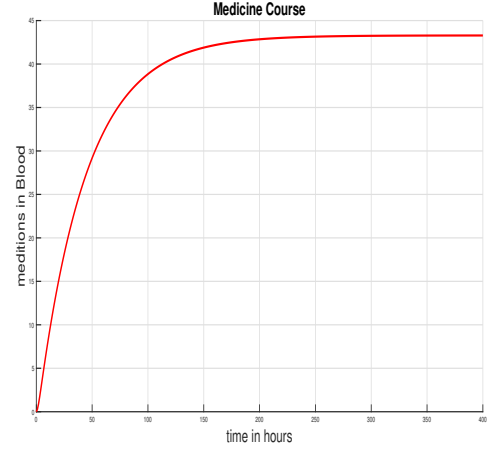


FIG. 7: $k_1 = 0.6931hr^{-1}$, $k_2 = 0.0231hr^{-1}$, $x_0 = 0 unit$, $y_0 = 0 unit$, $I = 1 unit$

The limiting value of $y(t)$ is **43.13722** units .

C. Plot the amount of medicine in a course of medicine

Fig. 6 shows the amount of medicine in GI track with respect to time.

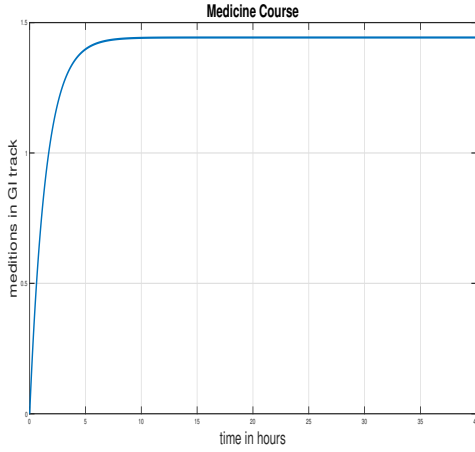


FIG. 6: $k_1 = 0.6931hr^{-1}$, $k_2 = 0.0231hr^{-1}$, $x_0 = 0 unit$, $y_0 = 0 unit$, $I = 1 unit$

The limiting value of $x(t)$ is **1.4467** units .

Fig. 8 shows the amount of medicine in GI track and blood with respect to time when $k_1=k_2$.

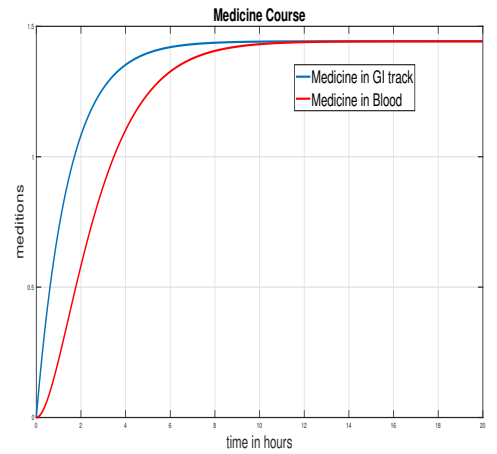


FIG. 8: $k_1 = k_2 = 0.6931hr^{-1}$, $x_0 = 0 unit$, $y_0 = 0 unit$, $I = 1 unit$

Fig. 9 shows the amount of medicine in GI track and blood with respect to time when $k_1=k_2$.

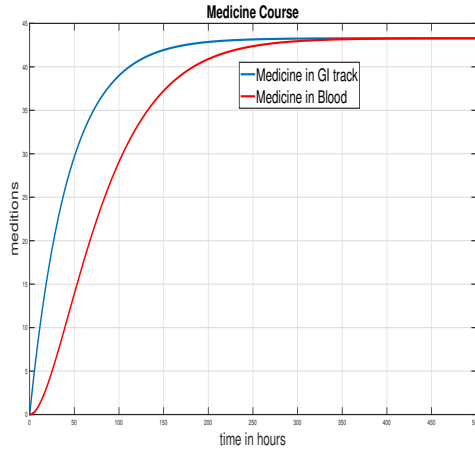


FIG. 9: $k_1 = k_2 = 0.0231hr^{-1}$, $x_0 = 0 \text{ unit}$, $y_0 = 0 \text{ unit}$, $I = 1 \text{ unit}$

IV. CONCLUSIONS

- From the results obtained, it is evident that the concentration of pollutants in the lake saturates to the value of C_{in} (inflow of pollutant) regardless of whether it is zero or non-zero.
- In the case of a single dose of medicine, the medicine concentration in the gastrointestinal (GI) tract exponentially decreases and converges to zero as t approaches infinity. Meanwhile, the medicine concentration in the blood initially rises to its peak, then exponentially decreases and converges to zero. The convergence time depends on the values of K_1 and K_2 .
- In the case of a course of medicine, the concentrations of medicine in the blood and GI tract both converge to the same value for $K_1 = K_2 = k$. Here, the convergence time depends on the value of K .